



GOLGI TENDON ORGAN REFLEX (GTO)

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OUTLINE

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- II) GOLGI TENDON ORGAN (GTO)
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I) ANATOMY OF THE SKELETAL MUSCLE

- Skeletal muscle is made up of two types of muscle fibers:
 - **Extrafusal muscle fibers**
 - Generate movement
 - Attach to tendons
 - **Intrafusal muscle fibers**
- Tendons are collagen fibrils which attach skeletal muscle to bone
- Weaved in and around these fibrils is the **Golgi Tendon Organ (GTO)**
 - A type of sensory receptor located within a capsule
 - The sensory fibers are called **Type 1b fibers**, which are afferent nerves that send signals to the CNS

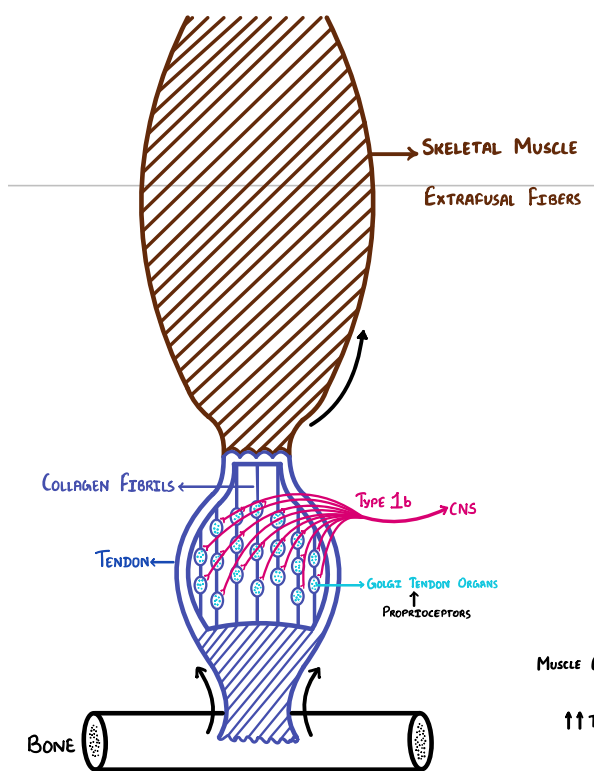


Figure 1. Skeletal Muscle

II) GOLGI TENDON ORGAN (GTO)

- Whenever a muscle contracts, it shortens and pulls on the tendon
- The tendon will pull on a bone and generate movement
- The **Golgi tendon organ (GTO)** can sense the degree of tension generated in the tendons from muscle contraction
 - They have **proprioceptors**, which are sensory receptors that send information about the position of muscles, joints, and ligaments
- When the Golgi tendon organs are stimulated, it activates the Type 1b sensory/afferent fibers

(A) ACTIVATION OF SENSORY FIBERS

- At the ends of the sensory fibers are mechanically-gated ion channels
 - They are sensitive to mechanical stimulation, such as the tension detected within a tendon
- **In the presence of tension**, the following happen:
 - the GTO is stimulated
 - the mechanically-gated ion channels within the membrane of the type 1b fibers opens, allowing extracellular cations, mainly sodium, to enter
- **When the cations enter**, it makes the interior of the cell more positive
 - The voltage-gated ion channels along the axon membrane are sensitive to a threshold potential, a specific voltage at which the channels open
 - These channels open and
 - the sodium ions flush in very strongly,
 - generating an action potential that
 - propagates down the axon towards the CNS/spinal cord

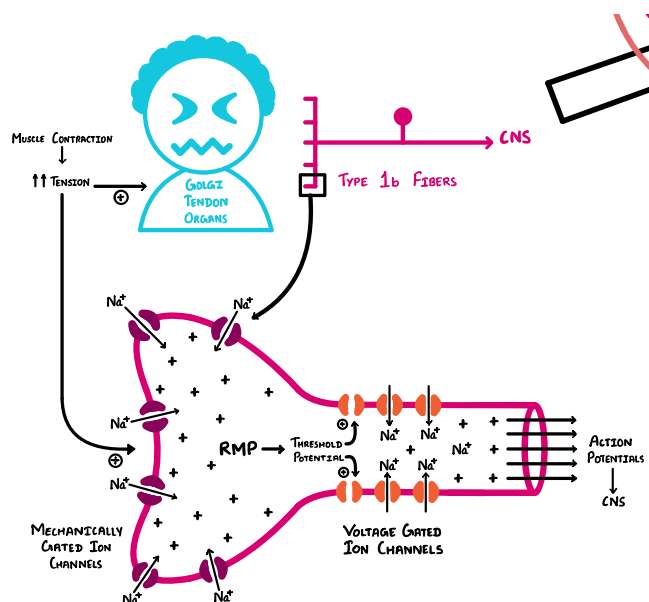


Figure 2. Activation of Type 1b Fibers



(B) GOLGI TENDON REFLEX

- Also known as the **inverse myotatic reflex**
- **Golgi Tendon Reflex Pathway**
 - Muscle contraction
 - → tendon is pulled
 - → tension is generated within the tendon
 - → **GTO** is activated
 - → GTO stimulates **Type 1b sensory neurons**
 - → action potential travels to the **dorsal root ganglion**
 - → action potential travels to the **posterior grey horn**
- From there, the **afferent nerve will bifurcate** and synapse onto two types of interneurons

(1) Inhibitory Interneuron

- The inhibitory interneuron **synapses** with a motor neuron on the **anterior grey horn**, which exits the spinal cord to innervate skeletal muscles
- When muscle contraction is excessive, there is greater risk of tendon avulsion; thus, the inhibitory interneuron is stimulated
- Example:
 - When the biceps muscle contracts excessively, the GTO detects the increased tension in the tendon
 - The type 1b sensory fibers are stimulated, and they activate the inhibitory interneuron
 - The inhibitory interneuron then stimulates the **alpha motor neuron** which innervates the biceps muscle, signaling it to relax (consequently reducing the tension)
 - The neurotransmitter used for this interneuron is **Glycine**, it generates **Inhibitory Post Synaptic Potentials (I.P.S.P.)**
 - Makes negative ions rush in or positive ions rush out (makes the inside of the cell really negative or hyperpolarized)

(2) Stimulatory Interneuron

- The stimulatory interneuron also **synapses** with a motor neuron on the **anterior grey horn**, which exits the spinal cord to innervate skeletal muscles
- The stimulatory interneuron stimulates the motor neurons which innervate the **antagonistic muscle**
- Example:
 - When the biceps muscle contracts excessively, the GTO reflex stimulates it to relax via the inhibitory interneuron
 - In addition to this action, the antagonistic muscle (i.e., triceps) needs to contract to pull the forearm in the opposite direction, to **stabilize the joint**
 - Thus, the **stimulatory interneuron** stimulates the **alpha motor neuron** of the triceps muscle, causing it to contract
 - The neurotransmitter used for stimulatory interneuron is **Glutamate**, it generates **Excitatory Post Synaptic Potentials (E.P.S.P.)**
 - Makes positive ions rush in or positive ions rush out (activating the cell and generating action potential (**Depolarize**))

Summary

- When there is excessive contraction of a muscle, the GTO reflex causes:
 - Inhibition of contracting muscle
 - Stimulation of antagonistic muscle
- For this reason, the GTO reflex is also referred to as "**reciprocal activation**" and "**autogenic inhibition**"

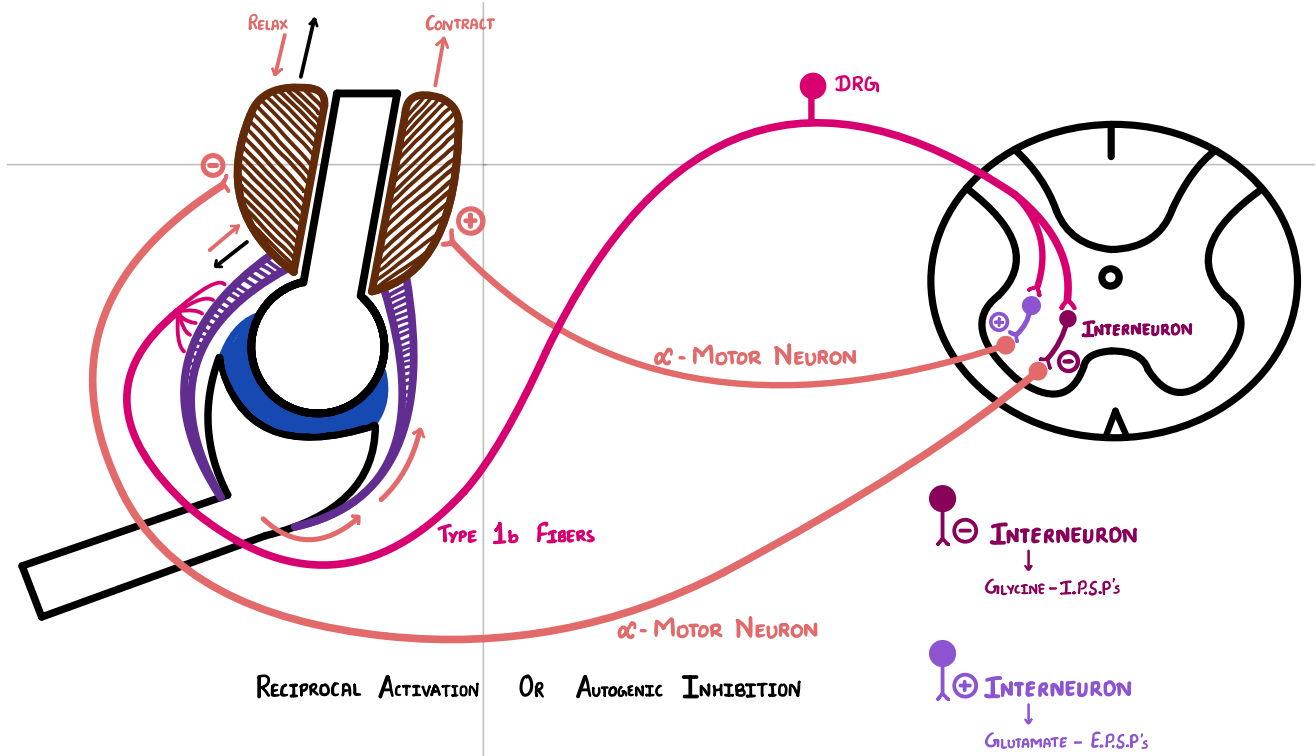


Figure 3. GTO Reflex

(C) NEUROTRANSMITTERS

- The inhibitory neurotransmitter is **glycine**
 - Generates inhibitory postsynaptic potentials, which cause negative ions to rush in or positive ions to rush out (hyperpolarization of the cell membrane)

- The stimulatory neurotransmitter is **glutamate**
 - Generates excitatory postsynaptic potentials, which causes cations to rush in the cells



III) REVIEW QUESTIONS

1) Which of the following is FALSE about the Golgi Tendon Organ?

- a. The GTO is located within the tendons of skeletal muscles.
- b. The GTO is stimulated whenever tension decreases in the tendon.
- c. The GTO stimulates Type 1b sensory fibers.
- d. The GTO reflex is mediated by two interneurons.

2) The Golgi Tendon Reflex is also known as the following, EXCEPT:

- a. Inverse myotatic reflex
- b. Autogenic inhibition
- c. Reciprocal inhibition
- d. Reciprocal activation

3) What type of ion channels are present along the axon of the Type 1b sensory fibers?

- a. Mechanically-gated ion channels
- b. Voltage-gated ion channels
- c. Ligand-gated ion channels
- d. Pressure-gated ion channels

4) Assuming the triceps muscle contracts excessively, which of the following is TRUE?

- a. The inhibitory interneuron is stimulated, and the biceps muscle relaxes
- b. The stimulatory interneuron is stimulated, and the triceps muscle contracts
- c. The inhibitory interneuron is inhibited, and the triceps muscle relaxes
- d. The stimulatory interneuron is stimulated, and the biceps muscle contracts

5) Which of the following statements is FALSE?

- a. Both the inhibitory and stimulatory interneurons are activated during excess muscle contraction
- b. The stimulatory interneuron synapses with the alpha motor neuron innervating the antagonistic muscle.
- c. The GTO reflex is also known as "reciprocal inhibition".
- d. The inhibitory neurotransmitter is glycine

[CHECK YOUR ANSWERS](#)

IV) REFERENCES

